

RAJAT SAVDEKAR

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EDUCATION

University of Southern California, Los Angeles, CA

Aug 2024 – May 2026

Master of Science in Computer Science

GPA: 3.63

Coursework: Algorithms, Database Systems, Machine Learning, Web Technology, VR / XR Development

University of Mumbai, Mumbai, India

Aug 2019 – May 2023

Bachelor of Engineering in Computer Engineering

CGPA: 9.4/10

Coursework: Operating Systems, Computer Networks, Distributed Systems, Natural Language Processing

SKILLS

Languages: Python, TypeScript, JavaScript, C++, C#, SQL, Java

AI & Agents: OpenAI API, LangChain, RAG, vector search, FAISS, PyTorch, model quantization

Backend & APIs: FastAPI, Flask, Node.js, REST APIs, WebSockets

Data & Storage: PostgreSQL, MongoDB, Redis, LevelDB, Yjs CRDTs

Cloud & Infrastructure: AWS, Docker, Terraform, GitHub Actions, CI/CD

Frontend & Spatial: React, Next.js, Unity, Meta Quest, Tailwind CSS

EXPERIENCE

Software Developer

Aug 2026 – Present

Easley-Dunn Productions, Inc. – Mediverse

Torrance, CA

- Engineered an **edge-AI pipeline** for junior doctor surgical training, cutting scenario feedback from 22 to 4 minutes by keeping a 4-bit quantized 7B model resident on a local server to bypass the Quest 1's memory limits.
- Optimized 6 core surgical tool modules for severely constrained hardware (4 GB RAM), automating **CI/CD testing** to prevent performance regressions and maintain a locked 72 FPS.

Teaching Assistant

May 2025 – Jul 2025

USC Summer Programs

Los Angeles, CA

- Mentored 150 students in **Python, Github, Debugging, Postman and REST APIs**, running daily code reviews to guide 20 teams from spec to hardware-integrated demos.

Software Engineer

May 2023 – May 2024

Jalgaon Fruit Sales Cooperative Ltd (JFSS)

Jalgaon, India

- Doubled **peak harvest write throughput** by architecting an isolated write path that eliminated the 10–15 minute MyISAM table-lock freezes stalling the platform every season, sustaining growth from \$100K to \$200K in transaction volume.
- Decomposed a **legacy PHP monolith** into modular Python Flask microservices on PostgreSQL, running both paths in parallel with cron-based state sync until cutover, cutting query latency 40% across 1,000+ daily truck transactions.
- Delivered a **web procurement platform** with REST APIs and live analytics dashboards, migrating 150 daily clerks across 5 warehouses off manual logbooks.

PROJECTS

MockPad – Real-Time Collaborative Code Editor

[\[Live\]](#) [\[GitHub\]](#)

Node.js, React, WebSockets, Yjs CRDTs, OpenAI Whisper, MongoDB, LevelDB

May 2026

- Supported **100 real users** with zero edit conflicts by architecting a real-time collaborative layer using Yjs CRDTs and WebSockets, restoring concurrent sessions cleanly upon reconnect.
- Sustained 100% uptime on a \$6/month droplet by engineering an **LRU eviction policy** that offloads idle rooms to LevelDB.

Caliberate – LLM Code-Evaluation Sandbox

[\[GitHub\]](#)

Python, FastAPI, PostgreSQL, Docker, GitHub Actions

Aug 2026

- Scaled concurrent model evaluations without race conditions by distributing **isolated Docker workloads** through a PostgreSQL SKIP LOCKED queue, ensuring safe execution and lease-based crash recovery.
- Eliminated manual review bottlenecks by wiring the sandbox into **GitHub Actions CI** to auto-reject failing LLM patches.

Celestia VR – Spatial Planetarium & AI Guide

[\[Demo\]](#) [\[Presentation\]](#) [\[GitHub\]](#)

Unity, C#, Meta Quest 3, Python, LangChain, RAG

Apr 2026

- Delivered instant contextual audio feedback for user queries by bridging a Meta Quest voice frontend with a Python RAG **pipeline** to retrieve and synthesize answers from an indexed stellar database.
- Maximized immersion by replacing hardware controllers with native dynamic hand-tracking and speech controls.

RESEARCH & HACKATHONS

Publications. *Texture Feature Analysis Using GLCM Matrix* a MATLAB feature-extraction tool since cited by 10+ papers in IEEE and Springer volumes. [\[GitHub\]](#)

Aura – Apple visionOS Accessibility Assistant. 2nd place of 40+ teams, USC 24-hour hackathon, Aug 2025.

[\[Demo\]](#)